



- appropriate service agent (weather, news, or joke).
- Each service agent generates a response, such as a weather forecast, news update, or joke.
- The system logs the time taken to process each query and the final response.
- The output shows the final response (query, category, and generated response) for each user query.

A framework for future AI systems

Multi-agent architectures that enable adaptive learning, dynamic decision-making, and frictionless integration into varied ecosystems are the future of AI. Next-generation agentic AI architecture is composed of five integrated layers that enable the system's overall function.

The Input Layer collects data from user interactions and real-time feeds to enable the system to be pre-empted with actionable intelligence. The Agent Orchestration Layer enables adaptive task management, collaborative actions, and performance monitoring. At each of these layers, agents have developed distinctive functions, such as planning and execution, self-scoring and optimising performance, employment of tools and domain-specific work, or continually learning and improving knowledge models. The Data Storage and Retrieval Layer receives centralised and distributed stores, optimised information access vector stores, and context-enriched knowledge graphs supporting context-based reasoning. In the Output Layer, AI insights are translated into individualised, context-aware outputs and continuously update system knowledge bases. The Service Layer provides enhanced AI capability across all platforms with cognitive recommendations and governance-compliance assurance.

This design also includes safeguards for safe, compliant, and responsible use of AI.

AI models designed in this way can keep a check on any bias, enable fairness and safety, and ensure compliance with regulations, ensuring the system remains strong in the long run. Also, partnership AI models enhance overall functionality by enabling interaction with external systems. **END** 🐧

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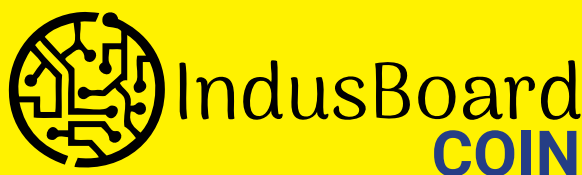
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